Zero-Shot & Unsupervised Phishing URL Intelligence Pipeline

ML & AI Security | Unsupervised Learning | Team Size: 3 | Duration: 8 Weeks

A detailed project proposal and operational plan for an explainable phishing detection architecture that combines high-dimensional vector embeddings, structured Chain-of-Thought (CoT) LLM reasoning, and prompt-injection defense mechanisms.

1. Abstract

Traditional supervised phishing classifiers depend heavily on static, labeled datasets that fail to generalize against novel adversarial infrastructure and rapid distribution shifts.

This project implements an unsupervised, explainable phishing detection architecture that combines high-dimensional vector embeddings, structured Chain-of-Thought (CoT) LLM reasoning, and prompt-injection defense mechanisms. The system operates without labeled training data by isolating structural anomalies mathematically and using constrained generative models to explain threat vectors.

To ensure complete modularity, the work is strictly decoupled into three independent technical tracks unified via a standardized JSON interface.

2. System Architecture & Team Division

The architecture is strictly separated into three independent modules, ensuring parallel development and clean integration.

Track 1: Vector Space Anomaly Detection (Unsupervised Learning)

- **Core Function:** Transforming raw, unparsed URL topologies into dense vector spaces.
- **Features:** Extracts token-level, structural, and lexical features (entropy, subdomain depth, path variance).
- **Modeling:** Uses open-source transformer models (e.g., all-MiniLM-L6-v2) to generate vector representations.
- **Indexing:** Indexes trusted root domains (Tranco/Cisco Umbrella) into a local vector database (FAISS/ChromaDB).
- **Scoring:** Computes an unsupervised Out-of-Distribution (OOD) anomaly score via k-NN and cosine distance.

Track 2: Agentic Threat Engine (Explainability & Reasoning)

- **Core Function:** Zero-shot semantic analysis and human-readable threat attribution.
- **Methodology:** Implements Chain-of-Thought (CoT) prompts to analyze misleading apex domains, brand impersonation, and homoglyphs.
- **Constraints:** Applies strict schema validation (Pydantic / Instructor) for deterministic, machine-readable JSON outputs.
- **Performance:** Evaluates latency and inference efficiency across local and hosted model backends.

Track 3: Adversarial Guardrails & LLMOps (Security)

- **Core Function:** Hardening the system against evasion and evaluating pipeline reliability.
- **Defense:** Mitigates Indirect Prompt Injection attacks (OWASP LLM Top 10) embedded inside URL parameters and encoded payloads.
- **Benchmarking:** Curates a verified evaluation set of benign, known phishing, and adversarial URLs.
- **Metrics & UI:** Builds an automated evaluation harness (Precision, Recall, F1) and an integrated reporting dashboard.

3. Feasibility & Resource Assessment

8 Weeks	Advanced	High
TIMELINE	DIFFICULTY	EFFORT LEVEL

Timeline: This project is aggressively scoped for 8 weeks. The strict decoupling of tracks allows the 3-member team to operate in parallel for the first 5 weeks before merging systems.

Difficulty Assessment: This project sits at the bleeding edge of AI and Cybersecurity. The primary challenges involve tuning unsupervised k-NN thresholds, constraining non-deterministic LLMs to rigid JSON outputs, and proactively red-teaming against Indirect Prompt Injections.

Effort Allocation: Each of the three members acts as a Domain Lead (approximately 15–20 hours/week per person). Weeks 6 and 7 will require intense collaborative debugging to ensure perfect data handoffs between the separate pipelines.

4. Operational Plan of Action

To execute the 8-week roadmap effectively, strict ownership has been established. The workflow distributes tasks evenly across Machine Learning, AI Engineering, and Security Operations.

Phase (Week)	Member 1 (ML / Data) Track 1 Lead	Member 2 (AI / Backend) Track 2 Lead	Member 3 (SecOps / QA) Track 3 Lead
Foundation (W1)	Set up Tranco Top 10k list; Define input data schemas.	Define JSON interchange schemas; Set up repository.	Draft threat taxonomy; Research OWASP LLM Top 10.
Foundation (W2)	Parse URL topologies; Generate baseline embeddings (MiniLM).	Draft initial zero-shot CoT prompts; Configure Pydantic.	Assemble initial benign, malicious, and injected benchmark URLs.
Core Build (W3)	Build embedding pipeline; Test vector distance metrics.	Implement CoT for typosquatting & brand impersonation.	Build decoding/sanitization engines for payload interception.
Core Build (W4)			Finalize the 60-sample benchmark dataset.

Phase (Week)	Member 1 (ML / Data) Track 1 Lead	Member 2 (AI / Backend) Track 2 Lead	Member 3 (SecOps / QA) Track 3 Lead
	Store Tranco vectors in local DB; Validate Tranco distances.	Enforce schema constraints; Handle API/ local model routing.	
Tuning (W5)	Calibrate decision boundaries and anomaly thresholds.	Refine prompts for edge cases (URL shorteners, dynamic routing).	Run evaluation harness on mock outputs to test pipelines.
Integration (W6)	*(Collaborative)* Connect Raw URL → Anomaly Scoring.	*(Collaborative)* Connect Input Sanitization → LLM Engine.	*(Collaborative)* Build main inspection interface / dashboard.
Validation (W7)	Analyze false positives/ negatives; Optimize latency.	Debug schema validation failures under heavy load.	Execute full benchmark; Stress-test prompt injections; Calculate metrics.
Wrap-up (W8)	Write methodology for vector scoring and DB architecture.	Write documentation on prompt design and JSON schemas.	Finalize evaluation report (F1-score, evasion rates).

> **CRUCIAL ADVICE FOR WEEK 1**
>
> Before writing a single line of machine learning code, all three members must agree on and hard-code the exact **JSON schemas** that will pass between Track 1, Track 2, and Track 3. If Member 1's anomaly output doesn't perfectly match what Member 3's sanitizer expects, the integration phase (Week 6) will bottleneck and fail.

5. Expected Deliverables

- **Modular Codebase:** A reproducible Python repository with zero tight-coupling dependencies across tracks.

- **Vector Database Index:** An unsupervised index (FAISS/ChromaDB) for real-time lexical anomaly scoring.
- **Agentic Reasoning Engine:** A constrained LLM pipeline providing structured, mathematically backed explainable threat attribution.
- **Adversarial Guardrails:** A robust security module capable of detecting and intercepting indirect prompt injections.
- **Comprehensive Report:** An evaluation document detailing system accuracy (Precision/Recall/F1), false positive rates, and security boundaries.